

Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering
CSE 3102, Lab No. 10 , Lab Work

E-Commerce Platform: Database Schema, Sample Data, SQL Queries

Scenario

You are tasked with designing and querying a database for an online E-commerce platform. The platform sells products across various categories, tracks customer orders, manages product inventory, and collects customer reviews.

Task 1: Database and Table Creation

Create a database called “EvalyDB”. Then create the following tables with the specified fields and constraints:

Customers

Field	Type	Constraints
customer_id	SERIAL	PRIMARY KEY
username	VARCHAR(50)	UNIQUE, NOT NULL
email	VARCHAR(50)	UNIQUE, NOT NULL
password	VARCHAR(50)	NOT NULL

Categories

Field	Type	Constraints
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category_id	SERIAL	PRIMARY KEY
name	VARCHAR(50)	NOT NULL

Products

Field	Type	Constraints
product_id	SERIAL	PRIMARY KEY
name	VARCHAR(100)	NOT NULL
price	DECIMAL(10,2)	NOT NULL
stock_quantity	INT	NOT NULL
category_id	INT	FOREIGN KEY REFERENCES Categories(category_id)

Orders

Field	Type	Constraints
order_id	SERIAL	PRIMARY KEY
customer_id	INT	FOREIGN KEY REFERENCES Customers(customer_id)
order_date	DATE	NOT NULL
status	VARCHAR(50)	NOT NULL

OrderDetails

Field	Type	Constraints
order_id	INT	FOREIGN KEY REFERENCES Orders(order_id)

product_id	INT	FOREIGN KEY REFERENCES Products(product_id)
quantity	INT	NOT NULL
PRIMARY KEY (order_id, product_id)		

Reviews

Field	Type	Constraints
review_id	SERIAL	PRIMARY KEY
customer_id	INT	FOREIGN KEY REFERENCES Customers(customer_id)
product_id	INT	FOREIGN KEY REFERENCES Products(product_id)
rating	INT	CHECK (rating >= 1 AND rating <= 5)
review_text	TEXT	

Task 2: Sample Data Insertion

Categories

category_id	name
1	Electronics
2	Clothing
3	Home & Garden

Products

product_id	name	price	stock_quantity	category_id
1	Laptop	1200.00	10	1
2	T-Shirt	20.00	50	2
3	Blender	80.00	15	3

Customers

customer_id	username	email	password
1	alice	alice@email.com	pass123
2	bob	bob@email.com	pass456
3	charlie	charlie@email.com	pass789

Orders

order_id	customer_id	order_date	status
1	1	2025-06-01	Delivered
2	2	2025-06-02	Shipped
3	1	2025-06-03	Pending

OrderDetails

order_id	product_id	quantity
1	1	1
1	2	2
2	3	1
3	2	1

Reviews

review_id	customer_id	product_id	rating	review_text
1	1	1	5	Excellent laptop!
2	2	2	4	Good quality shirt.
3	3	3	3	Works fine.

Task 3: SQL Queries and Example Results

a) How many products are there in each category?

```
SELECT c.name AS category, COUNT(p.product_id) AS product_count
FROM Categories c
LEFT JOIN Products p ON c.category_id = p.category_id
GROUP BY c.category_id, c.name;
```

category	product_count
Electronics	1
Clothing	1
Home & Garden	1

b) List the usernames and emails of customers who have made more than 1 order.

```
SELECT cu.username, cu.email, COUNT(o.order_id) AS order_count
FROM Customers cu
JOIN Orders o ON cu.customer_id = o.customer_id
GROUP BY cu.customer_id, cu.username, cu.email
HAVING COUNT(o.order_id) > 1;
```

username	email	order_count
alice	alice@email.com	2

c) What is the average rating for products in the 'Electronics' category?

```
SELECT AVG(r.rating) AS avg_rating
FROM Reviews r
JOIN Products p ON r.product_id = p.product_id
JOIN Categories c ON p.category_id = c.category_id
WHERE c.name = 'Electronics';
```

avg_rating
5.0

d) How many orders are in each status (Pending, Shipped, Delivered)?

```
SELECT status, COUNT(order_id) AS order_count
FROM Orders
GROUP BY status;
```

status	order_count
Delivered	1
Shipped	1
Pending	1

e) Which product has the highest stock quantity?

```
SELECT name, stock_quantity
FROM Products
ORDER BY stock_quantity DESC
LIMIT 1;
```

name	stock_quantity
T-Shirt	50

f) What is the total revenue generated from the 'Electronics' category?

```
SELECT SUM(od.quantity * p.price) AS total_revenue
FROM OrderDetails od
JOIN Products p ON od.product_id = p.product_id
JOIN Categories c ON p.category_id = c.category_id
WHERE c.name = 'Electronics';
```

total_revenue
1200.00

g) List the top 2 most ordered products.

```
SELECT p.name, SUM(od.quantity) AS total_ordered
FROM OrderDetails od
JOIN Products p ON od.product_id = p.product_id
GROUP BY p.product_id, p.name
ORDER BY total_ordered DESC
LIMIT 2;
```

name	total_ordered
T-Shirt	3
Laptop	1

h) Which customers have never left a review?

```
SELECT cu.username, cu.email
FROM Customers cu
LEFT JOIN Reviews r ON cu.customer_id = r.customer_id
WHERE r.review_id IS NULL;
```

username	email
(none)	(none)

(In this sample, all customers have left a review.)

i) What is the average order value for each customer?

```
SELECT cu.username, AVG(order_total) AS avg_order_value
FROM (
  SELECT o.order_id, o.customer_id, SUM(od.quantity * p.price) AS order_total
  FROM Orders o
  JOIN OrderDetails od ON o.order_id = od.order_id
  JOIN Products p ON od.product_id = p.product_id
  GROUP BY o.order_id, o.customer_id
) AS order_sums
JOIN Customers cu ON order_sums.customer_id = cu.customer_id
GROUP BY cu.customer_id, cu.username;
```


username	avg_order_value
alice	820.00
bob	80.00

j) Which products have never been ordered?

```
SELECT p.name
FROM Products p
LEFT JOIN OrderDetails od ON p.product_id = od.product_id
WHERE od.order_id IS NULL;
```

name
(none)

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